

Natural Language Processing with Transformers

AI Learning Hub - Comprehensive Course Guide

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NLP Fundamentals

Natural Language Processing enables computers to understand human language.

Text Preprocessing Pipeline:

1. Tokenization: Splitting text into words or subwords.
2. Stop Word Removal: Removing common words (the, is, at).
3. Stemming and Lemmatization: Reducing words to root forms.
4. Vectorization: Bag of Words, TF-IDF, Word2Vec, GloVe.

Word Embeddings map words to dense vectors where semantically similar words are close in vector space. Word2Vec uses Skip-gram or CBOW architectures. GloVe uses co-occurrence statistics from a corpus.

The Transformer Architecture

Transformers, introduced in 'Attention Is All You Need' (2017), revolutionized NLP.

Key Components:

- Self-Attention Mechanism: Computes attention scores between all positions in a sequence. Q (Query), K (Key), V (Value) matrices. $\text{Attention}(Q,K,V) = \text{softmax}(QK^T/\sqrt{d_k})V$
- Multi-Head Attention: Runs multiple attention operations in parallel.
- Positional Encoding: Adds position information since transformers lack recurrence.
- Feed-Forward Networks: Applied independently to each position.
- Layer Normalization and Residual Connections for training stability.

The encoder processes input, the decoder generates output autoregressively.

BERT, GPT, and Modern LLMs

BERT (Bidirectional Encoder Representations from Transformers):

- Pre-trained on Masked Language Modeling and Next Sentence Prediction.
- Bidirectional context understanding.
- Fine-tuned for classification, NER, question answering.

GPT (Generative Pre-trained Transformer):

- Autoregressive model trained to predict next token.
- GPT-2, GPT-3, GPT-4 show scaling laws: larger models perform better.
- In-context learning: few-shot, zero-shot capabilities.

Modern trends: instruction tuning, RLHF (Reinforcement Learning from Human Feedback), mixture of experts, retrieval-augmented generation (RAG).